

Title: More Efficient Clinical Trials: The Center for Bayesian Adaptive Trials - CBAT

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/Applicant]
Gui Liberali	EUR	RSM	liberali@rsm.nl	Tenured	Researcher	Main Lead
Eric Boersma	EMC	Cardiology	h.boersma@erasmusmc.nl	Tenured	Researcher	Lead
Hester Lingsma	EMC	Public Health	h.lingsma@erasmusmc.nl	Tenured	Researcher	Applicant
Diederik Dippel	EMC	Neurology	d.dippel@erasmusmc.nl	Tenured	Mix	Applicant
Jasper Brugts	EMC	Cardiology	j.brugts@erasmusmc.nl	Tenured	Mix	Applicant
Carin Uyl-de Groot	EUR	ESHPM	uyl@eshpm.eur.nl	Tenured	Researcher	Applicant
Joost van Rosmalen	EMC	Epidemiology	j.vanrosmalen@erasmusmc.nl	Tenured	Researcher	Applicant
Sebastian Gabel	EUR	RSM	gabel@rsm.nl	Tenure track	Researcher	Applicant
Alina Ferecatu	EUR	RSM	ferecatu@rsm.nl	Tenure track	Researcher	Applicant
David van Klaveren	EMC	Public Health	d.vanklaveren@erasmusmc.nl	Tenured	Researcher	Applicant
Erik van Zwet	LUMC	Clinical Epidemiology	e.w.van_zwet@lumc.nl	Tenured	Researcher	Applicant
Rolf Groenwold	LUMC	Clinical Epidemiology	r.h.h.groenwold@lumc.nl	Tenured	Researcher	Applicant

Corresponding Lead: Gui Liberali

Planned start date of seed funding: 01/12/2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Tertiary Convergence theme (if applicable): Choose an item.

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: *Bayesian, statistics, machine learning, clinical trials*

1.b Summary (max 300 words)

This flagship has the objective of improving how clinical trials are designed, monitored, and analysed, which will be achieved by developing the Center for Bayesian (Adaptive) Trials (CBAT). This will be a Center of knowledge and excellence in Bayesian and Bayesian Adaptive (BBA) clinical trials in medicine. The Center is designed around the research program described in this proposal. This program will develop frameworks, novel methods and guidelines that help researchers and trialists with the design, implementation, and monitoring of BBA clinical trials in medicine. These methods and practices will have important short- and long-term societal impact. First, BBA clinical trials have higher beneficence, as they reduce the number of patients exposed to harmful or inferior treatments. Second, BBA trials reduce the time-to-market and development costs of new medical

drugs and devices by reducing the total number of patients in trials. The Center will synthesize advances in clinical trial research, statistics, and operations research by identifying the elements in each field that can improve clinical trials in medicine. The core idea for this flagship program is based on two adaptive Bayesian methods that the applicants have developed and applied in two large scale trial data in cardiology recently as a proof-of-concept [2]. The main lead has previously applied Adaptive Bayesian methods in other settings, now published in non-medical journals [1,3,4,6]. The Flagship will further develop these methods, instruct others in their application, and apply them in real-time in forthcoming clinical trials. One of such trials is a novel registry-based Bayesian clinical trial to study the effectiveness of selenium and CoQ10 supplementation in heart failure, funded by the Heart Foundation. The research output will include methodological manuscripts (e.g. new BBA methods) and applied manuscripts (e.g. analyzing trials influenced by these methods) that can be used by scientists and research organisations.

1.c Public Summary (max 300 words)

Increasingly, the development and testing of medical drugs and devices are under pressure to reduce the length and cost of Phase III clinical trials *while also* reducing harm to patients. The pandemic made the balance between speed and statistical robustness in trials ever more salient and fervently discussed in major newspapers, TV programs, and social media by the general population, lawmakers, clinicians, and government officials. There is a massive need for faster and yet safer trials. This is becoming even more pressing in the E.U. and in the U.S. given long-term trends in medicine, such as the emergence of personalized medicine, and the recent changes in regulatory focus towards Bayesian trials [9]. Recent advances in statistics and machine learning in non-medical fields [1,3,4,6] resulted in sophisticated statistical methods that can make rapid and better use of patient information and responsibly adapt randomization rates. In this convergent flagship, we pooled together world-class researchers in medicine and applied machine learning to translate these breakthrough developments into methods that can deliver safer and faster clinical trials. The applicants have shown the potential of such methods using data from past trials [2], and now, together with PhD students, will further develop these methods and apply them in real-time in forthcoming clinical trials. One of such trials is a novel registry-based Bayesian clinical trial to study the effectiveness of selenium and CoQ10 supplementation in heart failure, funded by the Heart Foundation. The research will be hosted in a multi-campus research center: the Center for Bayesian (Adaptive) Trials (CBAT). The center is designed around 4 long-term research lines. The center will also have strong societal and valorization arm, expressed in yearly events designed for, and open to clinicians, regulators, and pharma companies.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e (word count: 2483)

2.a Contribution to H&T mission and fit with H&T theme(s)

To improve how clinical trials are designed, monitored, and analysed, we will set up the Center for Bayesian (Adaptive) Trials (CBAT), a center of knowledge and excellence in Bayesian and Bayesian Adaptive (BBA) clinical trials in medicine. CBAT will develop frameworks, novel methods, and guidelines that help researchers and trialists design, implement, and monitor BBA clinical trials.

Because CBAT is a methodological Flagship, it offers the opportunity to transcend beyond its own borders and enact change on a higher level. The methodological advances that CBAT will develop can in turn affect other Flagships working on substantive healthcare problems. By transcending disciplines and diseases, CBAT embodies the philosophy of Convergence.

CBAT contributes to all aspects of the Convergence Health & Technology mission:

- Life-long health and well-being, from pre-birth to end of life
To ensure life-long health in the future, we need to speed up and personalize medical development. CBAT supports this with BBA trial developments that reduce the time, money, and human effort required to test new treatments. This method is applicable to trials of new drugs, new medical devices or interventions, and post-market surveillance.
- Socio-economic equality in health(care)
Since BBA trials cost less than current randomized controlled trials (RCTs), they reduce development costs, leading to cheaper and more accessible healthcare.
- Prevention and early diagnostics to proactively maintain health
Application of the BBA method would benefit the market journey of preventative treatments as well.
- Individual-tailored precise medical treatment for all
As the BBA method allows the conduct of trials with significantly fewer people than trials based on the frequentist philosophy, disease stratification and unique personal aspects can be considered. In some cases, e.g. for rare diseases with low population sizes, trials would simply be impossible without the use of BBA.

CBAT also fits with all Convergence Health & Technology Themes:

- Human Centered Technology and AI for Health
The CBAT objectives and work programme, using AI methods and focusing on personalized healthcare, fit strongly with this theme. CBAT aims to use human-centered AI and technological advances.
- Technology Supported Transitions in Health(care)
CBAT will provide methodological and statistical support for further advances in medicine. With its sophisticated statistical approaches and use of big data and AI, it will accelerate healthcare developments.
- Improving Health Journeys
BBA clinical trials will improve health journeys drastically by reducing the burden of patients during trials and by ensuring more effective drugs/treatments that can quickly be marketed.
- Fundamentals of Health & Disease
In the future, more and more information will be used in the ever-learning BBA model, so it can also test fundamentals of health and disease, such as specific genetic or molecular causes.

2.b Synergy between disciplines

The CBAT program merges approaches and insights from four main disciplines:

Statistics

Adaptive Bayesian methods originate in statistics and operations research. Members of the CBAT team have developed these methods for >10 years, applying them in non-medical situations

(business and marketing) [1,3]. Now, sophisticated Bayesian Adaptive Trials enable per-patient adaptation as opposed to two-stage clinical trials [2].

Computer science

Recent developments of algorithms in reinforcement learning and in multi-armed bandits (MAB) allow for real-time updating and prediction, automated action, and continuous learning. This enables continuous adaptation and BBA methods. During a trial, the algorithm decides in real-time what treatment the next patient will be assigned to, based on all available patient information up to that moment. BBA methods and training will be heavily inspired by these developments.

Medicine

The synergy between statistics/machine learning and medicine in the CBAT team ensures that method development closely follows the needs of trialists. We have shown the feasibility and benefit of our methods in an analysis of existing RCT data [3], which will further support the adoption of Bayesian methods in medicine. Furthermore, CBAT is closely related (JB, EB, EvZ, RG are co-investigators) with the Heart4Data consortium that will run a Dutch national heart failure registry and a registry-based Bayesian adaptive trial of selenium and CoQ10 supplementation in heart failure patients.

Regulatory/ethics

The use of BBA methods in clinical research can improve an important regulatory/ethics dilemma: current clinical trials are not the most efficient or safest. However, given the strong oversight and regulatory bodies in the medical field, changes to the status quo also require regulatory/ethical recognition. To reach our goal of adoption by clinicians, we therefore incorporated regulatory and ethical angles in our team and work plan.

2.c Long term sustainability plan

Feasibility of the program

Methodologically, we believe that the CBAT program is very feasible:

1. The core principles are based on >10 years of work of the applicants with Adaptive Bayesian methods in statistics and operations research [1, 3, 4, 6]
2. The Flagship leads have extended and applied one of these methods to data from two cardiology trials and shown its efficacy as a proof-of-concept [2]. This study is in the process of publication.
3. The Flagship leads and scientific advisor Prof. John Hauser are in the process of creating a company (Sairona B.V.) that will provide services in this field, which will help translate research into practice. They have applied to the NWO Venture Challenge and have been chosen to participate.

Success of the program also depends upon the willingness of regulators and clinicians to adopt the methodology. We believe this aspect is feasible as well:

1. Top medical journals have acknowledged the potential and necessity for the use of Bayesian (adaptive) methods in clinical trials [5,7,8].
2. Clinical investigators and sponsors acknowledge the added value of a Bayesian approach. The REMAP-CAP initiative is exemplary in this respect. The Heart4Data project, recently funded by the Dutch Heart Foundation, also includes a BBA clinical trial.

3. Pharmaceutical companies have shown interest in the methodology. One pharmaceutical company (Servier) provided the data for our proof-of-concept paper [2].
4. The FDA is aware of the potential of Bayesian methods in clinical trials and is open to their use [9]. EMA is having ongoing discussions along the same lines. Our methods are an extension of this.
5. Cardialysis, a contract research organization that has been running trials for more than 30 years, has expressed its interest in collaboration to expand their portfolio with BBA methodology.

Long-term intellectual sustainability

The mission of CBAT is to become the go-to destination for researchers and trialists interested in Bayesian trials in the EU. To ensure the longevity of the Center, our work programme is organized in broad, semi-permanent, multi-year research lines, rather than work packages. Our work plan (elaborated in 3c) aims to establish a robust Center with long-term intellectual sustainability.

Long-term financial sustainability

Long-term financial sustainability will be secured via the following routes:

- Public funding such as NWO/ZonMw (e.g. OTP, Perspectief) and Horizon Europe (e.g. EIC). CBAT aligns very well with (inter)national research agendas (see 3d) and will therefore be able to apply to many relevant calls. Our team has an excellent track-record of receiving such funding (see 4a).
- Collaborations with industry, such as pharmaceutical and medical device companies. These stakeholders stand to benefit immensely from the development of our methodology. We therefore plan to collaborate closely with them on future clinical trials and aim for them to contribute to CBAT with more structural funding as well. We are currently discussing collaboration options with Cardialysis and the Dutch Cardiovascular Alliance, who are interested in applying our methodology.
- Funding has been obtained from the Hartstichting (Heart Foundation) to run a registry-based BBA clinical trial to study the effectiveness of selenium and CoQ10 supplementation in heart failure (see budget).

Risks and mitigation measures

The main risk in completing the goals of CBAT is that existing adaptive trial methods are not yet widely used by trialists, meaning our method requires a culture change and a change in the workflow of clinical trials.

Although we are confident that regulators and clinicians will adopt the methodology (see 'Feasibility' above), we have also taken measures to minimize this risk:

- CBAT is a close collaboration between methodologists and clinicians. This ensures that methodological developments follow the needs of clinical environments and allows for quick uptake of the methodology.
- This uptake takes the form of an initial Bayesian adaptive clinical trial. The results of this trial, together with our results from analyzing previous RCTs [2], will be used to convince more and more stakeholders of the benefits of BBA methods.
- We have a thorough communication, dissemination, and collaboration plan (see 2d) that will help to spread our work.
- We devote part of our work plan to ethical and regulatory questions and aim for a collaboration with relevant regulatory bodies. One of our research lines is focused on regulation, tackling issues such as design and registration and monitoring and stakeholders.

2.d Visibility, attracting talents and new partners

Creating visibility for CBAT

Academia and knowledge institutes

The scientific world will be kept up to date with our research through peer-reviewed publications and preprints, online available teaching material, an e-book with protocols and methods, presentations at scientific conferences, and development of software packages, models and workflows. We also reserved budget for visiting scientists who are drawn to our Center from around the world.

Industry, regulators, and medical field

We believe that our initial work in showing the effectiveness of Bayesian adaptive trials will go a long way in creating visibility with the relevant industries, regulatory bodies, and the medical field. To boost this further, we will organize a yearly impact event where representatives from all these stakeholders will be invited (approximately 100 participants). At this event, we will present our latest results, discuss the directions that the research can take, gain feedback on utilization options, and organize panels to debate relevant current topics. Stakeholders will get access to the latest ideas in BBA methods and a broad network of researchers in the field. Attendance will be free to sponsors of CBAT (see 'Attracting Partners' below). We expect to benefit from future private partnerships and additional project ideas through these networks. Cardialysis, mentioned above, has expressed keen interest in CBAT and in joining its events, discussions, and projects.

Society

Our research and results will also be disseminated through a Center website, social media (e.g. Twitter, Instagram, Blogs, LinkedIn), and traditional media (popular magazines, newspapers, interviews on radio and television, and newsletters).

Engaging students and early career researchers (ECRs) in interdisciplinary projects

To ensure the impact and long-term sustainability of the Center, there will be a strong training and teaching arm within CBAT. The main activities are:

Education activities

We will exchange and develop course material for the MSc and BSc level. The consortium actively teaches BSc and MSc courses and is committed to introducing novel techniques and insights into its curricula. In fact, lead applicant Gui Liberali has already developed and implemented a course on the first CBAT method, which students from all over the EU are welcome to attend. In 2022, the course will take place in August. Professor Liberali already offered this course twice, under a different name, in 2021 and in 2020, always with a minimum of 20 students. This course is part of a Research Master's (in Business Data Science) which Prof. Liberali co-developed. We envision that CBAT will eventually lead to the creation of a specialized Research Master's.

Research assistantships and thesis projects

The type of work done in the Center is very data-intensive, which creates opportunities for research assistantships. In this way, MSc and BSc students can assist the ECRs working in the Center and gain valuable experience. Analysis of RCT data is already a common thesis project. With some guidance, students can be trained to analyze adaptive RCT data.

Training and collaboration

PhD students will have the opportunity to receive training in specialized topics, general skills, responsible research practices, dissemination, and more. We will further encourage PhDs and postdocs to spend three months either at one of the other participating institutes, with the 2nd supervisor, or at another institute. This immerses them in another research culture and the consortium will benefit from stronger coherence. We also aim to hold monthly research seminars, where ECRs, faculty, and visiting researchers can present and discuss their work. All this will encourage sharing and create opportunities for joint (interdisciplinary) projects.

Interactions with industry

As outlined above, CBAT aims for strong links with industry. In addition to the yearly impact event, where ECRs will be involved, we will also explore the possibilities of traineeships for industry-minded PhDs.

Attracting talents, partners, and funding

CBAT aims to become the go-to Center for expertise on Bayesian (adaptive) trials. This will naturally attract talented people who want to work on this topic and/or achieve impact in the field of clinical trials. We further aim to be an attractive Center to work at because of the strong cooperative and interdisciplinary nature, the opportunities for personal and professional development (outlined above), and the strong links with industry. PhD students and postdocs will furthermore each receive a personal travel and training budget of €5k for participation in (inter)national conferences, summer schools, and academic events.

To attract partners and funding, we will offer sponsorship opportunities to stakeholders. We will create an advisory council with industry partners. They will be able to advise the Center on their main concerns, which they would like to see researched in the coming years. They will also be invited to our yearly impact event. With this event, we hope to attract more partners and funding.

Our consortium already has a wide network of both academic and industry partners (see section 4c). This will help us to attract and select both talent and partners.

2.e Long term societal impact

CBAT has the potential to revolutionize RCTs, and thereby the way that the efficacy of new treatments is demonstrated. The impact of this necessary innovation cannot be understated, as it will result in countless lives, money, and time saved. To achieve the below-mentioned impacts, we aim to follow the impact pathways detailed in Figure 1 and the strategies for visibility outlined in section 2d.

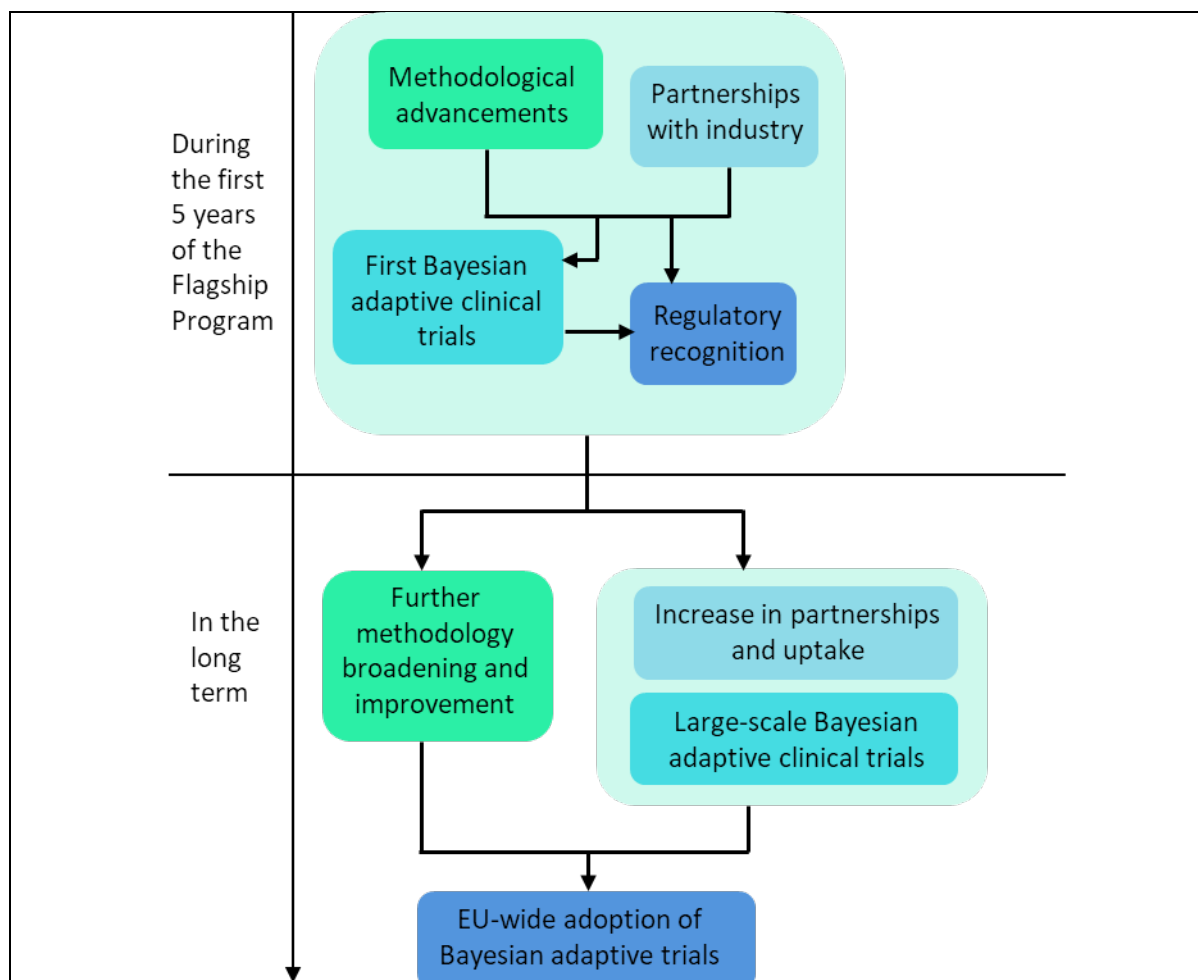


Figure 1. Impact pathways. Methodological advances and partnerships with industry will lead to the first Bayesian adaptive trials and, with the help of these trials, to regulatory recognition (note that regulations do allow the use of Bayesian methods, but no official guidelines or recommendations have been published about Bayesian adaptive trials). These outcomes pave the way for further methodology broadening and improvement, as well as increased partnerships and uptake and large-scale Bayesian adaptive clinical trials. The ultimate driver of societal impact will be EU-wide adoption of Bayesian adaptive trials.

Quantitatively, we believe CBAT has the potential to reach the following impact in the long term (~10 years):

- 25% average reduction of necessary sample size in trials.
- 2-20% of lives or major complications of trial participants saved (depending on similarity of medications tested).
- 10-20% increase in clinical trials instead of observational studies for rare diseases and personalized treatments.
- Effective treatments will reach the market and change guidelines on average 6-12 months earlier.
- 25% cost reduction due to the possibility to include less patients in each trial, quicker elimination of ineffective new treatments, and faster time-to-market of effective new treatments.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d (word count: 1997)

3.a Ambitions

Through setting up the CBAT expertise center, we aim to increase the **efficiency of clinical research** through the methodology of **Bayesian adaptive trials**. Using advanced mathematics based on multi-armed bandits and stochastic process theory, **these methods dynamically alter the randomization rate after each patient, reducing the number of patients exposed to harmful treatments**. Furthermore, these methods can decide after each included patient **whether a trial should be continued or stopped** for efficacy, safety, or futility, obviating the need to specify time points for interim analyses before the trial starts. This approach **reduces the sample size** of a clinical trial without detrimental impact on statistical power. Not only are these methods more efficient, they are also more ethical than a standard trial without adaptive allocation ratio or early stopping. These gains in efficiency are sorely needed in **precision medicine** and disease areas where patient populations tend to be small. Another CBAT ambition is to play a leading role in the development and dissemination of Bayesian adaptive trials, so we aim to become the go-to destination for researchers and trialists interested in Bayesian adaptive trials in the EU in the next 5-10 years.

In this Flagship project we translate advanced methodology from marketing research to the medical field [1,3,4,6]. These methods are further developed and evaluated using expertise from applied medicine, medical decision making, and public health. We will thus bring knowledge from various fields together, in line with the objectives of Convergence. In CBAT, researchers from different faculties/universities will collaborate.

A challenge of the Flagship is that trial practitioners, regulatory agencies, and other stakeholders may be reluctant to consider alternative methodology, especially because the standard two-arm RCT with fixed equal randomization ratio has been a cornerstone of clinical research since the 1950s. We aim to avoid this by including both clinical researchers, pharmaceutical companies and regulatory agencies (i.e. EMA) from the start. Furthermore, one of the proposed research lines is on the regulatory aspects of adaptive trials, to help all stakeholders, including regulators, transition to this new paradigm.

Another potential challenge is the complexity of the methodology, especially where the methodology needs to be extended to consider both safety and efficacy of treatments. Therefore, our project may be considered a high-risk, high-gain project. However, our team has long-running expertise in Bayesian adaptive trials, and the methodology has been proven to work. Furthermore, in addition to the publications on the potential of Bayesian methods and multi-armed bandits in clinical trials [5,7,8], the analysis performed by the lead applicants on clinical trial data [2] illustrate how much more efficient BBA trials are (in terms of time and number of patients) compared to the RCT golden standard.

3.b Aims and feasibility

The main aims of the program are:

- Develop a mature and comprehensive methodology for Bayesian adaptive clinical trials.

- Make the adaptive trial designs easy to use for trialists and other clinical researchers, by providing advice, training, and by developing freely-available software tools.
- Develop a center of expertise for Bayesian adaptive trials through which the research is performed, and that acts as a conduit for disseminating knowledge on this methodology.

One of the pillars of CBAT is to create a sustainable research Center with a view to the future. Fostering talent from students, including PhD, MSc, and BSc students, therefore plays an essential role in our strategy. There will be many opportunities for MSc/BSc students to take part in research internships and conduct research at our Center for their theses. The development of PhD students is described in section 2.d.

Feasibility

The basic methodology has previously been applied in marketing research, several applicants are experts in this area, and we have a proof of concept of the methodology in medicine [2]. Extending and evaluating the methodology, using simulation studies and real-life case studies, can therefore be done soon after the initiation of the project in tandem.

The requested funding suffices for approximately 6 PhD candidates and 6 Postdocs. Each research line is quasi-permanent and has been designed such that there is work for approximately 4-7 research papers. Supervision of PhD candidates is done by the applicants, for which no funding is requested. Additional funding is currently available through our collaboration with the Heart4Data consortium. In addition, we anticipate that additional funding will become available from future grants (see 2.c) and commercial parties such as Cardialysis.

Most of the research will be methodological, tested in live trials in cooperation with partners such as the Heart4Data consortium and CROs such as Cardialysis. The research teams will occasionally also perform simulations using past trial data. Data access will be obtained from collaborations being discussed with CROs, HOVON, other existing networks (see 4c), and online data sharing platforms such as YODA. We have experience with such data requests, and no problems are anticipated here.

In this project, we bring together expertise from various scientific fields. The applicants combine expertise from marketing science, cardiology, biostatistics, public health, and medical decision making and all are (leading) experts in their respective fields.

Possible risks and mitigations:

- Difficulty of extending methodology to more complex/realistic outcomes, such as survival data. For some statistical models, such as Cox regression for survival data, the methodology may turn out to be complex or even infeasible. However, our methodological success does not depend on one approach and this flexibility allows us to mitigate methodological risks.
- Difficulty of getting buy-in from pharmaceutical partners. The long-term success of CBAT depends on funding from pharmaceutical companies and other commercial partners. In the first phase it will be difficult to get buy-in from these companies and there is some risk that long-term funding will not pan out. To address this risk, we will approach commercial partners early. In fact, we have already approached some commercial partners (e.g. Cardialysis) that are interested.

3.c Work plan

Research line 1: Decision-analytic framework for adaptive trials

Recent developments in Bayesian adaptive trials allow for per-patient, continuous adaptation, as opposed to two-stage designs based on interim analysis. We will develop a broad framework for continuously adaptive trials that extend these methods and set the path for future developments. The framework will (a) explicitly account for early stopping for futility, safety, and efficacy; (b) integrate subgroup analysis; (c) handle biomarkers and surrogate endpoints, and (d) allow the introduction of new arms into the continuous-trials literature.

Research line 2: Time and Dynamics in Adaptive Trials

We will extend the methods to time-to-event (i.e. survival) data and integrate hazard models into the multi-armed bandit framework for adaptive clinical trials.

We will also deal with non-stationarity and drift due to incremental changes in treatment and setting. We will study this from a within-trial monitoring and a post-trial analysis point of view.

Research line 3: Cohort and historical data in adaptive trials

Historical data often provide valuable information. Where available, we can combine data of suitably chosen historical controls with data of the current trial to increase statistical power and precision, and reduce sample size requirements. We will develop methodology to include the adaptive use of historical data in the Bayesian adaptive clinical trial framework.

More specifically, we will combine historical with current data in such a way that the historical data are fully used when they are similar to the current data and that the historical data are down-weighted or even discarded when there are substantial differences. We will develop methods to make the randomization ratio and the sample size of the new trial adaptive, considering the similarity of the historical and randomized controls.

Research line 4: Regulatory issues

In this research line, we will investigate and contribute to the regulatory framework regarding adaptive trials, in collaboration with the FDA and EMA and in line with the spirit of ongoing discussion at these organizations. Questions include, but are not limited to:

- How to design and register protocols for continuously adaptive trials, including sample size, power, and other statistical and safety concerns.
- How to improve the role of stakeholders and processes to guarantee the merits of the blind process regarding stopping, compliance, and relevant trade-offs.
- How can we interact and demonstrate structure and validity of trials that will be proposed using the adaptive methodology?

Activities that will be performed across research lines:

- **Implement the continuously adaptive method in new trials**
We will design and implement an RCT that implements the Bayesian adaptive methodology. Our center of expertise will be an example and serve as a source of information so that other clinical researchers can easily set up these trials. This includes weakly-coupled collaboration, such as providing guidance and advice to clinicians and pharmaceutical companies that want to use our methods, and strongly-coupled collaborations, where the CBAT researchers and PhD students are responsible for the main statistical decisions and protocol design.
- **Develop, publish, and maintain R packages to facilitate the adoption of new trials**
We will also develop software tools (e.g. a package in R) that enables other researchers to easily design, set up, and analyze new clinical trials using the Bayesian adaptive methodology.

- **Present the results and on-going developments in CBAT's yearly workshop together with Pharma companies and regulators.**

This will generate insights that inspire our researchers, facilitate funding efforts, and promote transparency and accountability to CBA.

Required facilities

Most of the research will be methodological. We need good computing facilities (e.g. a cluster) may to perform simulation studies (e.g. high capacity servers, 30k euro per year). To run a trial, we need software tools to do randomization in real time, i.e. something like the Alea service of the Clinical Trial Center.

Data and infrastructure plan

In the first years, we will collect no patient data, but we will evaluate the methodology using data of existing clinical trials. The data will be used solely for the purposes of developing and testing the BBA methods and publication of the findings. Researchers will analyze the data on an anonymized database behind the firewall of the Erasmus MC. Aggregated results will be published upon approval by trial sponsors.

3.d Alignment of the program with relevant national and international research agendas

The objective of CBAT is to promote safer and more ethical trials by developing the BBA trial methodology and to promote and co-execute full Bayesian (adaptive) clinical trials. This objective aligns with national and international research agendas:

Dutch National Science Agenda (NWA)

The approach of CBAT will be a real *game changer* – significantly reducing the time, cost and human effort required to bring new medications to clinical practice. CBAT's research program is therefore well-linked to the NWA Routes and Cluster Questions aimed at healthcare breakthroughs and those dealing with big data. The methodology and insights gained from CBAT will enable and/or accelerate progress of these questions and serve as a basis for further development and solutions.

The most relevant NWA questions to which CBAT will contribute are:

- **CQ094:** How do we improve the quality of healthcare and keep it affordable?
- **CQ095:** How can healthcare, for example by using biomarkers, be more focused on a person's uniqueness?
- **CQ098:** How can we better translate breakthroughs in fundamental biomedical research into the development of new medicines?
- **CQ102:** How can we develop new medicines and treatments to stay as vital and healthy as possible?
- **CQ105:** How can big data and technological innovation (e-health) contribute to healthcare?
- **CQ112:** Big data: can we use large data sets and their collection to realize values, generate insights and obtain answers?

European Priorities and Missions

One of the priorities of the European Commission for the coming years is "A Europe fit for the Digital Age". Its aim is to make the digital technology transformation work for people and businesses. As explained in section 2.a, this matches closely with the objectives of CBAT.

With its innovation of the way clinical trials are run, CBAT will also contribute to the EU priority "Beating Cancer" and the associated Horizon Europe Cancer Mission ". This is because adoption of the CBAT methods will reduce the burden of patients during medical trials and allow effective treatments to reach consumers faster. CBAT methods would work both on the 'prevention' and 'cure' side of the cancer mission.

4. Consortium (25%) max 2000 words for 4.a to 4.d (word count: 1951)

4.a Track record of the consortium members

Our consortium consists of 10 researchers from the Erasmus Medical Center and the Erasmus University Rotterdam, and 2 researchers from Leiden University Medical Center, as shown in Table 1 (arranged by PhD year; leads in bold).

Table 1: Consortium members

Name	PhD year	Expertise	Grants, prizes, awards
Prof. dr. ir. H. Boersma	1998	Biostatistics, clinical epidemiology of cardiovascular diseases	Various Dutch and European grants (e.g. ZonMw, NWO, Heart

			Foundation, FP7). Total €20.7m since PhD.
Prof. dr. G. Liberali	2006	Bayesian adaptive trials and machine learning	Marie Curie grant, NWO/STW take-off grant, Top Talent Researcher Erasmus School of Economics, ERIM Fellow, BDS Fellow, Academic Director of ECDA.
Prof. dr. D. Dippel	1994	Neurology, new statistical methods in neurology	Numerous grants for efficiency research, from e.g. Dutch Heart Foundation, Dutch Brain Foundation, pharmaceutical companies. Winkler medal (for MR CLEAN).
Prof. dr. C. Uyl-de Groot	1995	Regulatory aspects of clinical studies	Numerous national and international grants, incl. 3 ZonMw grants (2 as project leader), 4 H2020 grants, and 10 grants with funding from private parties. Total €7.26m in the past 10 years.
Dr. E. van Zwet	1999	Medical statistics, Bayesian methods in trials	Post-doctoral fellowship, visiting scholarships.
Dr. J. Brugts	2009	Cardiology, epidemiology	Various research and clinical grants to study treatment effects in cardiology, from e.g. ZonMw, Dutch Heart Foundation, Dutch Society of Cardiology, Zorginstituut, and medical technology companies. In total €8.4m in the past 10y as PI.
Dr. J. van Rosmalen	2009	Biostatistics, Bayesian methods for dynamic borrowing of clinical and analysis of longitudinal data	Numerous grants for biostatistical projects and consultations (from e.g. Sanquin Blood Supply, Health~Holland, Department of Public Health at Erasmus MC).
Prof. dr. R. Groenwold	2009	Causal modelling, efficiency assessment methods	ZonMw (Veni, Vidi), LUMC Fellowship.
Prof. dr. H. Lingsma	2010	Quantitative methods, prediction modelling, causal inference from observational data	Veni, 2 fellowships, 5 Erasmus MC doelmatigheid (3 as PI), 2 ZonMw, 1 NWA, 5 grants from NGOs (4 as PI), and others. In total €5.5m since PhD.
Dr. A. Ferecatu	2014	Multi-armed bandits and adaptive trials	Various research scholarships and doctoral fellowships.
Dr. D. van Klaveren	2017	Prediction modelling, model validation and individual treatment decision making.	Three PCORI (patient-centered outcomes research institute USA) grants, 1 as PI. One ZonMw (as PI), 1 NWO, and 1 Horizon Europe grant.
Dr. S. Gabel	2019	Deep learning and machine learning	Two scholarships, UNC Kenan-Flagler research grant, EHI Science Prize: Best Dissertation.

Academic research

Our Center harbors excellence in all the different relevant disciplines and levels, as exemplified by the publications, (personal) grants, and awards achieved by its researchers. Our team members have published in key journals for their disciplines, such as Lancet, British Medical Journal, New England Journal of Medicine, JAMA, Circulation, European Heart Journal, Journal of the American College of Cardiology, Management Science, Marketing Science, International Journal of Research in Marketing, and the European Journal of Operations Research. They are highly cited and those at mid-career level or higher are considered experts in their field. They also review for various (high impact) journals. Our early career researchers are already exhibiting excellence, as evidenced by their research scholarships, grants, and fellowships.

Translation of scientific results into clinical application and societal impact

Our consortium members work closely together with clinical and industry partners. This is most evident from the grants they have received, which show many private sources of funding. For instance, Dr. van Rosmalen regularly performs statistical consultations, which often play an important role in the development of new clinical treatments. Other team members, such as Prof. Dippel and Dr. Brugts, frequently research the effectiveness and cost-efficiency of treatment methods. Prof. Dippel led the MR CLEAN study, which revolutionized ischemic stroke treatment and received much media attention worldwide.

Several members also serve on scientific advisory boards, committees, and councils that aim to translate scientific results into clinical practice or societal impact. As an example, both Prof. Boersma and Prof. Uyl-de Groot are members of the Health Council of the Netherlands (Gezondheidsraad). In this capacity, they advise the Dutch ministers and Parliament, which maximizes the impact of health(care) research on society.

From the methodological side, an excellent example of the translation of scientific results into societal impact is the technical paper by Prof. Liberali and colleagues [1]. This paper was featured by the Sloan Management Review, a marketing science journal which explains technical papers to a lay audience consisting of e.g. managers and law makers.

BSc and MSc supervision and engagement in research projects

All consortium members have ample experience supervising BSc and MSc students as part of their thesis research and for research internships. Each team member supervises on average 5 MSc and BSc students per year for research projects. Further, they are involved in mentoring talented students and fostering research skills in MSc and BSc students. For example, Prof. Lingsma received a grant to organize statistics consultations for Medicine students working on their MSc thesis.

Ability to attract external funding

A summary of the grants obtained can be found in the table above. Naturally, most consortium members have experience with obtaining prestigious grants such as H2020, ZonMw, NWO, and Marie Curie. For each of these awards, the researcher must be in the top 10% of their peer group. This is an indication that future grant proposals will be successful as well, strengthening the financial security of CBAT.

It is important to note the various grants obtained from public-private partnerships and private parties such as pharmaceutical companies, hospitals, companies, and non-profit organizations. Not only are they evidence of the translation of scientific results into clinical practice and societal

impact, but they also represent a crucial revenue stream for CBAT. Since an important aim of CBAT is the practical application of the developed methodology, cooperation with and funding from partners such as these is something we strive for. The track-record of attracting such external funding is evident.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The main aims of the program are:

- Develop a mature and comprehensive methodology for Bayesian adaptive clinical trials, with the features described in the work plan.
- Make the adaptive trial designs easy to use for trialists and other clinical researchers, by providing advice, training and by developing software tools that will be made freely available.
- Develop a center of expertise for Bayesian adaptive trials through which the research is performed, and that acts a conduit for disseminating knowledge on this methodology.

To reach these objectives, we have put together a consortium of excellent team members with expertise in Bayesian adaptive methods, machine learning, biostatistics, clinical trials, and regulatory issues. Specific expertise for each member can be found in Table 1.

Prof. Liberali, Dr. van Rosmalen, Dr. Gabel and Dr. Ferecatu, with external support of Prof. Hauser, are our experts on Bayesian methods in combination with machine learning. They will work on the methodological development of the Bayesian adaptive clinical trials.

Further methodological expertise comes from Dr. van Klaveren, Prof. van Zwet and Prof. Groenwold. They are specialized in biostatistics, including Bayesian methods.

Prof. Boersma and Prof. Lingsma are our clinical trial experts. They are exceptionally knowledgeable about the ins and outs of clinical trial design, implementation, and execution.

Clinical experts are Prof. Dippel and Dr. Brugts. They will keep the methodological advancements focused on the needs and barriers that trialists and clinical researchers experience.

Prof. Uyl-de Groot is our expert of regulatory issues that surround the adoption of new methodology in clinical practice. With her, our expertise Center for Bayesian Adaptive Trials is complete.

With the combination of methodology-driven and substantive trial experts from EMC and EUR, we believe we have put together an excellent consortium that will be able to complete CBAT's aims and objectives. Where beneficial, we have sought outside expertise, for instance from our collaborators from Heart4Data Prof. Groenwold and Dr. van Zwet. We further benefit from the experience of scientific advisor Prof. Hauser (MIT), the world expert on Bayesian adaptive trials in non-medical fields. We see opportunities for collaboration with TU Delft and will seek this in particular for research line 4.

4.c Involvement of partners

Our consortium members all have large research networks and work together with researchers and clinical partners in (inter)national projects, for instance Horizon Europe and ZonMw projects. These

collaborations will be used to get access to data for this Flagship. As an example, Prof. Uyl-de Groot is currently involved in Horizon Europe projects RETHRIM, T2Evolve, and CARAMBA.

Flagship lead Prof. Liberali works closely with MIT professor John Hauser, who is an expert on Bayesian adaptive methods and who will serve as a scientific advisor to CBAT. Together with the other lead, Prof. Boersma, they are in the process of creating a spin-off company (Sairona B.V.) which will provide services in the field of Bayesian Adaptive Trials. This will help translate research into practice.

Dr. Brugts is leading consortium Heart4Data, in which Prof. Broersma, Prof. Groenwold and Dr. van Zwet also participate. This project was recently funded (€1.2m) by the Dutch Heart Foundation. The project will culminate in a BBA clinical trial (€1m – in the process of being funded), studying the effectiveness of selenium and CoQ10 supplementation in heart failure. These external matching funds have been included in the budget.

Recently, lead applicants Prof. Boersma and Prof. Liberali have also been in contact with Cardialysis, a Contract Research Organization that has been running trials for more than 30 years. This company has expressed keen interest in CBAT and in joining its events, discussions, and projects.

Finally, our consortium members have contacts with various associations, societies, boards, and committees. In particular, our contacts with the Medical Evaluation Board (College ter Beoordeling van Geneesmiddelen), the European Health Association, and the European Society of Medicine Oncology will be helpful in the implementation of Bayesian adaptive trials.

4.d Governance, organization and collaboration strategy of the consortium

The Center is organized into four different research lines, each with a Research Lead and a Co Research Lead (which is essential in case of illness or a scientist leaving their position). These Research Leads will come together in the Steering Committee, which will be led by lead applicants Prof. Liberali and Prof. Boersma who will serve as Directors.

The steering committee will come together approximately 3-4 times per year to discuss progress, research directions, funding, etc. Of course, more informal discussions and collaboration will take place throughout the Flagship and beyond. The day-to-day running of the Center, including the preparation of yearly reports to Convergence, will be done by the Center Manager, for whom hiring will begin as soon as the Flagship has been awarded funding.

Consortium members are connected to the Research Lines in the following way:

Research line 1: Decision-Analytic Framework for Adaptive Trials

Lead: Prof. Liberali

Co-lead: Dr. Brugts

Participants: Prof. Lingsma, Prof. Groenwold

Research line 2: Time and Dynamics in Adaptive Trials

Lead: Dr. van Klaveren

Co-lead: Dr. van Rosmalen

Co-lead: Dr. Gabel

Research line 3: Cohort and Historical Data in Adaptive Trials

Lead: Dr. van Rosmalen
Co-lead: Dr. van Zwet
Co-lead: Dr. Ferecatu

Research line 4: Regulatory Issues

Lead: Prof. Uyl-de Groot
Co-lead: Prof. Boersma
Participant: Prof. Dippel

To further the development of our ECRs (Dr. Ferecatu and Dr. Gabel) in a constructive way, we have chosen them to co-lead research lines 2 and 3 alongside more experienced colleagues.

Research line 1 provides the general framework that Research lines 2 and 3 will follow. Research line 4 will provide the legal and regulatory underpinning for the Flagship. Research lines 2 and 3 are highly methodological lines that will run in parallel. To facilitate inter-departmental cooperation, research lines 1, 2, and 3 will each have at least one PhD-student/Postdoc from EMC and one from RSM. Research line 4 will have at least one PhD-student/Postdoc from EMC and one from ESHPM.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

IV References

Budget formulier: Convergentie Flagship Call

Flagship:	Center for Bayesian Adaptive Trials
Flagship acronym:	CBAT
Naam Penvoerder:	
Startdatum en looptijd van het pr	Start date 01-12-2022, duration 5 years (60 months), 5-year end date: 30-11-2027
Totaal project budget:	

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																							
Erasmus MC		Center manager	0,4	1-12-2022	31-10-2027	Schaal 11	€ 5.584	€ 197.674	€ 8.376	€ 63.658	€ 25.128	€ 25.128	€ 25.128	€ 25.128	€ 50.256	€ 197.674	€ 0	€ 197.674				€ 197.674	
Erasmus MC		Postdoc	1,0	1-12-2022	30-11-2024	Schaal 8	€ 3.863	€ 139.068	€ 5.795	€ 69.534	€ 63.740	€ 69.534	€ 63.740	€ 69.534	€ 139.068	€ 139.068	€ 0	€ 139.068				€ 139.068	
Erasmus MC		Postdoc	1,0	1-4-2023	31-3-2025	Schaal 8	€ 3.863	€ 139.068		€ 52.151	€ 69.534	€ 17.384	€ 69.534	€ 139.068	€ 0	€ 139.068	€ 0	€ 139.068				€ 139.068	
Erasmus MC		PhD student	1,0	1-4-2023	31-3-2027	Schaal 5	€ 2.975	€ 214.200		€ 40.163	€ 53.550	€ 53.550	€ 53.550	€ 214.200	€ 0	€ 214.200	€ 0	€ 214.200				€ 214.200	
Erasmus MC		Postdoc	1,0	1-1-2025	31-12-2026	Schaal 8	€ 3.863	€ 139.068					€ 69.534	€ 139.068	€ 0	€ 139.068	€ 0	€ 139.068				€ 139.068	
Erasmus MC		PhD student	2,0	1-9-2023	31-8-2027	Schaal 5	€ 2.975	€ 428.400		€ 35.700	€ 107.100	€ 107.100	€ 107.100	€ 107.100	€ 71.400	€ 428.400	€ 0	€ 428.400				€ 428.400	
Erasmus MC		Postdoc	1,0	1-1-2026	31-12-2027	Schaal 8	€ 3.863	€ 139.068					€ 69.534	€ 139.068	€ 0	€ 139.068	€ 0	€ 139.068				€ 139.068	
Erasmus MC		PhD student	1,0	1-1-2024	31-12-2027	Schaal 5	€ 2.975	€ 214.200			€ 53.550	€ 53.550	€ 53.550	€ 53.550	€ 53.550	€ 214.200	€ 0	€ 214.200				€ 214.200	
Erasmus University Rotterdam		Postdoc	1,0	1-1-2023	31-12-2024	Schaal 8	€ 3.863	€ 139.068		€ 69.534	€ 69.534		€ 69.534	€ 139.068	€ 0	€ 139.068	€ 0	€ 139.068				€ 139.068	
Erasmus University Rotterdam		PhD student	1,0	1-4-2023	31-3-2027	Schaal 5	€ 2.975	€ 214.200		€ 40.163	€ 53.550	€ 53.550	€ 53.550	€ 53.550	€ 13.388	€ 214.200	€ 0	€ 214.200				€ 214.200	
Erasmus University Rotterdam		PhD student	1,0	1-1-2024	31-12-2027	Schaal 5	€ 2.975	€ 214.200			€ 53.550	€ 53.550	€ 53.550	€ 53.550	€ 53.550	€ 214.200	€ 0	€ 214.200				€ 214.200	
Erasmus University Rotterdam		Postdoc	1,0	1-1-2025	31-12-2026	Schaal 8	€ 3.863	€ 139.068					€ 69.534	€ 139.068	€ 0	€ 139.068	€ 0	€ 139.068				€ 139.068	
Organisatie 3							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Organisatie 4							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Ondernemingen																							
Onderneming 1							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Onderneming 2							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Onderneming 3							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Onderneming 4							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Overige (private) partners																							
Naam 1							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Naam 2							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Naam 3							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Naam 4							€ -	€ 0								€ 0	€ 0				€ 0	€ 0	
Totaal								€ 2.317.282	€ 14.171	€ 0	€ 370.901	€ 0	€ 549.236	€ 0	€ 502.880	€ 0	€ 555.030	€ 0	€ 325.065	€ 0	Totaal loonkosten	€ 2.317.282	

2.	Kosten van materialen en hulpmiddelen:								2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
	Partner die de kosten maakt	Materiaal / hulpmiddel					Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties																								
Erasmus MC		Clinical trial software. €10,000 per year x 2 years					€ 20.000							€ 10.000		€ 10.000,00					€ 20.000	€ 0	€ 20.000,00	
Erasmus MC		Valorization-related costs.€10,000 per year x 5 years					€ 50.000	€ -		€ 10.000,00		€ 10.000,00		€ 10.000		€ 10.000,00		€ 10.000,00		€ 10.000,00		€ 50.000,00	€ 0	€ 50.000,00
Organisatie 3																					€ 0	€ 0	€ 0,00	
Organisatie 4																					€ 0	€ 0	€ 0,00	
Ondernemingen																								
Onderneming 1																					€ 0	€ 0	€ 0,00	
Onderneming 2																					€ 0	€ 0	€ 0,00	
Onderneming 3																					€ 0	€ 0	€ 0,00	
Onderneming 4																					€ 0	€ 0	€ 0,00	
Overige (private) partners																								
Naam 1																					€ 0	€ 0	€ 0,00	
Naam 2																					€ 0	€ 0	€ 0,00	
Naam 3																					€ 0	€ 0	€ 0,00	
Naam 4																					€ 0	€ 0	€ 0,00	
Totaal								€ 70.000	€ 0	€ 0	€ 10.000	€ 0	€ 10.000	€ 0	€ 20.000	€ 0	€ 20.000	€ 0	€ 10.000		Totaal kosten materialen en hulpmiddelen	€ 70.000		

3.	Kosten van gebruik van machines en apparatuur					2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt		Machine / apparatuur		Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																				
Erasmus University Rotterdam		Computing facilities (servers)		€ 30.000	5	€ 150.000	€ 30.000,00		€ 30.000		€ 30.000,00		€ 30.000		€ 30.000,00			€ 150.000	€ 0	€ 150.000,00
Erasmus University Rotterdam		Laptops for new staff		€ 1.000	5	€ 5.000			€ 2.000		€ 2.000,00		€ 1.000					€ 5.000	€ 0	€ 5.000,00
Erasmus MC		Laptops for new staff		€ 1.000	9	€ 9.000	€ 2.000,00		€ 4.000		€ 1.000,00		€ 1.000		€ 1.000,00			€ 9.000	€ 0	€ 9.000,00
Organisatie 3						€ 0												€ 0	€ 0	€ 0,00
Organisatie 4						€ 0												€ 0	€ 0	€ 0,00
Ondernemingen																				
Onderneming 1						€ 0												€ 0	€ 0	€ 0,00
Onderneming 2						€ 0												€ 0	€ 0	€ 0,00
Onderneming 3						€ 0												€ 0	€ 0	€ 0,00
Onderneming 4						€ 0												€ 0	€ 0	€ 0,00
Overige (private) partners																				
Naam 1						€ 0												€ 0	€ 0	€ 0,00
Naam 2						€ 0												€ 0	€ 0	€ 0,00
Naam 3						€ 0												€ 0	€ 0	€ 0,00
Naam 4						€ 0												€ 0	€ 0	€ 0,00
Totaal						€ 164.000	€ 32.000	€ 0	€ 36.000	€ 0	€ 33.000	€ 0	€ 32.000	€ 0	€ 31.000	€ 0	€ 0	Totaal kosten machines en apparatuur		€ 164.000

4.

Aan derden verschuldigde kosten						2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
Partner die de kosten maakt		Naam derde, omschrijving kosten		Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																					
Erasmus MC		Yearly industry meeting catering. 100 attendees x€20 x 5 years		€ 10.000				€ 2.000		€ 2.000		€ 2.000		€ 2.000		€ 2.000		€ 10.000	€ 0	€ 10.000	
Erasmus MC		Budget for visiting researchers. 2 researchers per year x€2000		€ 20.000				€ 4.000		€ 4.000		€ 4.000		€ 4.000		€ 4.000		€ 20.000	€ 0	€ 20.000	
Erasmus MC		Trial costs		€ 1.200.000				€ 240.000		€ 240.000		€ 240.000		€ 240.000		€ 240.000		€ 1.200.000	€ 0	€ 1.200.000	
Erasmus University Rotterdam		Methodology workshops. 3 workshops x 30 attendees x€20		€ 1.800						€ 600		€ 600		€ 600				€ 1.800	€ 0	€ 1.800	
Organisatie 4																		€ 0	€ 0	€ 0	
Ondernemingen																					
Onderneming 1																		€ 0	€ 0	€ 0	
Onderneming 2																		€ 0	€ 0	€ 0	
Onderneming 3																		€ 0	€ 0	€ 0	
Onderneming 4																		€ 0	€ 0	€ 0	
Overige (private) partners																					
Naam 1																		€ 0	€ 0	€ 0	
Naam 2																		€ 0	€ 0	€ 0	
Naam 3																		€ 0	€ 0	€ 0	
Naam 4																		€ 0	€ 0	€ 0	
Totaal						€ 1.231.800	€ 0	€ 0	€ 246.000	€ 0	€ 246.600	€ 0	€ 246.600	€ 0	€ 246.600	€ 0	€ 246.000	€ 0	Totaal kosten derden		€ 1.231.800

5.

Publicatie, reis- en verblijfkosten						2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt		Naam derde, omschrijving kosten		Kosten		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																				
Erasmus MC		Travel budget for scientific staff €5000 x 13		€ 65.000	€ -			€ 12.500		€ 12.500		€ 12.500		€ 12.500		€ 15.000		€ 65.000	€ 0	€ 65.000,00
Erasmus MC		Open access publication costs. €6000 per year, starting in the 2nd year		€ 24.000	€ -			€ 0		€ 6.000		€ 6.000		€ 6.000		€ 6.000		€ 24.000	€ 0	€ 24.000,00
Erasmus University Rotterdam		Travel budget for scientific staff. €5000 x 9		€ 45.000	€ -			€ 8.000		€ 8.000		€ 8.000		€ 10.500		€ 10.500		€ 45.000	€ 0	€ 45.000,00
Erasmus University Rotterdam		Open access publication costs. €4000 per year, starting in the 2nd year		€ 16.000	€ -			€ 0		€ 4.000		€ 4.000		€ 4.000		€ 4.000		€ 16.000	€ 0	€ 16.000,00
Organisatie 4																		€ 0	€ 0	€ 0,00

6.

FINANCIERING - Bijdragen per partner per jaar

In Euro		2022		2023		2024		2025		2026		2027		Total		Total per partner
		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
1	Erasmus MC Flagship contribution	€ 18.000		€ 307.000		€ 300.000		€ 303.000		€ 300.000		€ 273.000		€ 1.501.000	€ 0	€ 1.501.000
2	Erasmus University Rotterdam Flagship contribution	€ 7.000		€ 85.000		€ 100.000		€ 97.000		€ 100.000		€ 76.500		€ 465.500	€ 0	€ 465.500
3	Organisatie 3													€ 0	€ 0	€ 0
...	Organisatie 4													€ 0	€ 0	€ 0
Totale bijdrage onderzoeksorganisaties		€ 25.000	€ 0	€ 392.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 349.500	€ 0	€ 1.966.500	€ 0	€ 1.966.500
Ondernemingen																
4	Various companies to be contacted for financial contributions							€ 32.000		€ 85.000		€ 68.000		€ 185.000	€ 0	€ 185.000
5	Onderneming 2									€ 0		€ 0		€ 0	€ 0	€ 0
6	Onderneming 3									€ 0		€ 0		€ 0	€ 0	€ 0
...	Onderneming 4									€ 0		€ 0		€ 0	€ 0	€ 0
Totale bijdrage ondernemingen		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 32.000	€ 0	€ 85.000	€ 0	€ 68.000	€ 0	€ 185.000	€ 0	€ 185.000
Overige (private) partners																
7	Hartstichting (Heart Foundation) current registration pro	€ 21.171		€ 291.401		€ 469.336		€ 0		€ 0		€ 0		€ 781.908	€ 0	€ 781.908
8	Hartstichting (Heart Foundation) future trial project							€ 399.980		€ 405.630		€ 194.064		€ 999.674	€ 0	€ 999.674
9	Naam 3									€ 0		€ 0		€ 0	€ 0	€ 0
...	Naam 4									€ 0		€ 0		€ 0	€ 0	€ 0
Totale overige bijdrage		€ 21.171	€ 0	€ 291.401	€ 0	€ 469.336	€ 0	€ 399.980	€ 0	€ 405.630	€ 0	€ 194.064	€ 0	€ 1.781.582	€ 0	€ 1.781.582
Totaal budget per jaar		in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind					Totaal
		€ 46.171	€ 0	€ 683.401	€ 0	€ 869.336	€ 0	€ 831.980	€ 0	€ 890.630	€ 0	€ 611.564		€ 3.933.082		€ 3.933.082
																Funding gap
																0
Totaal budget		€ 46.171		€ 683.401		€ 869.336		€ 831.980		€ 890.630		€ 611.564		€ 3.933.082		

Department of Marketing Management
Prof. dr. Stefano Puntoni
Room T10-21
E: spuntoni@rsm.nl

Subject: Convergence of Health and Technology initiative

Date
Our reference
Your reference

Rotterdam, 11 March 2022
SP/ab

Letter of support,

Gui Liberali, professor of digital marketing at the department of marketing management applied to the Convergence of Health and Technology initiative of Erasmus University (www.convergence.nl/call-for-flagship-programs). Together with a colleague from the department of Cardiology at Erasmus MC, Prof. Eric Boersma, they are submitting a proposal to the Flagship program, called "CBAT- Center for Bayesian Adaptive clinical Trials." In this proposal, they wish to promote safer and more ethical clinical trials by developing, applying, and disseminating novel optimization methods for trials.

This initiative is in line with the strategic goals of our department, such as strengthening the collaboration between EUR and Erasmus Medical Center, attracting research funds from outside of the department, and publication in high-impact journals.

Thus, we want to express our full support to this initiative. If you have any questions, please do not hesitate to contact me.

ValidSigned by Stefano Puntoni
on 11-03-2022

Stefano Puntoni
Head of the department of Marketing Management
Rotterdam School of Management
Erasmus University

References

1. Hauser, J.R., G.L. Urban, G. Liberali, and M. Braun. 2009. Website morphing. *Marketing Science* 28 (2): 202–224.
2. Liberali G, Boersma E, Lingsma H, Brugts J, Dippel D, Tijssen J, Hauser J. Adaptation of Bayesian Trials in the Presence of Interim Information. Working Paper. 2022
3. Liberali, G. G. & Ferecatu, AM. A., 10 Feb 2022, In: *Marketing Science*. Accepted. DOI <https://doi.org/10.1287/mksc.2021.1346>
4. Urban, G., Liberali, G., Bordley, R., Macdonald, E. & Hauser, J., 2014, In: *Marketing Science*. 33, 1, p. 27-46 20 p.
5. Schmidt C. The struggle to do no harm in clinical trials. *Nature*. 2017 Dec 21;552(7685): S74-S75. doi: 10.1038/d41586-017-08705-4. PMID: 29293235
6. Hauser, J.R., G. Liberali, and G.L. Urban. 2014. Website morphing 2.0: Switching costs, partial exposure, random exit, and when to morph. *Management Science* 60 (6): 1594–1616.
7. Angus D. Optimizing the Trade-off Between Learning and Doing in a Pandemic, *JAMA*. 2020;323(19):1895-1896. doi:10.1001/jama.2020.4984
8. Berry D. A Case for Bayesianism in Clinical Trials. *Stat Med*. 1993 Aug;12(15-16):1377-93; discussion 1395-1404. doi: 10.1002/sim.4780121504. PMID: 8248653.
9. FDA. Adaptive Design Clinical Trials for Drugs and Biologics Guidance for Industry Guidance Document, U.S. Department of Health and Human Services, Food and Drug Administration, Center for Drug Evaluation and Research, November 2019; FDA-2018-D-3124.